

MODULE-1

Session 4

(TASKS)

1. List all 6 phases of SDLC in order with one line about each.

Ans:

1. Requirement Analysis: -

Requirement Analysis is the first phase of the Software Development Life Cycle (SDLC). In this phase, developers, analysts, and clients discuss and gather all the requirements of the project to clearly understand what the software should do and what features it should include.

2. System Design: -

After the requirements are finalized, the System Design phase begins. In this phase, the overall architecture of the software, database structure, user interface, and technical specifications are planned to create a blueprint for development.

3. Development (Coding): -

The Development phase is where programmers write the source code based on the design documents. Different modules of the software are developed and integrated to create the complete application.

4. Testing: -

The Testing phase ensures that the software works correctly and is free from bugs and errors. Testers perform different types of testing, such as functional, performance, and security testing, to verify that the application meets the required standards.

5. Deployment (Implementation): -

Once testing is successfully completed, the software is deployed to the production environment where users can access and use it. This phase may also include installation, configuration, and user training.

6. Maintenance: -

Maintenance is the final and ongoing phase of SDLC. After the software is released, developers continue to fix bugs, improve performance, update features, and make changes according to user feedback and business requirements.

Session 4 (TASKS)

2. Write 3 differences between Agile and Waterfall models.

Ans:

Agile Model	Waterfall Model
Agile is an iterative and flexible software development model where the project is completed in small cycles called sprints.	Waterfall is a traditional and sequential model where each phase is completed before moving to the next phase.
In Agile, requirements can be modified or updated at any stage based on customer feedback and changing needs.	In Waterfall, requirements are usually fixed at the beginning, and making changes later can be difficult and costly.
Agile involves continuous collaboration between developers, testers, and customers throughout the project.	Waterfall has limited customer involvement after the initial planning stage, and the final product is usually delivered at the end of the project.

3. Name the 3 main Scrum roles and write one line about the responsibility of each.

Ans:

1. Scrum Master: -

The Scrum Master is responsible for ensuring that the Scrum framework is properly followed by the team. They remove obstacles that affect the team's productivity, organize Scrum meetings, and help the team work efficiently.

2. Product Owner: -

The Product Owner acts as the representative of the customer and stakeholders. They create and manage the product backlog, prioritize tasks according to business value, and ensure that the development team works on the most important features first.

3. Development Team: -

The Development Team consists of software developers, testers, designers, and other technical professionals who work together to build, test, and deliver a functional product increment at the end of every sprint.

- These three Scrum roles work together to improve collaboration, increase productivity, and deliver high-quality software.

Session 4 (TASKS)

4. What is a Sprint in Scrum? Mention its typical duration.

Ans:

- A **Sprint** is a fixed and time-boxed period in the Scrum methodology during which the development team completes a selected set of tasks from the product backlog. The main objective of a sprint is to produce a working and potentially releasable version of the software.
- At the beginning of a sprint, the team conducts Sprint Planning to decide which tasks will be completed. During the sprint, team members collaborate, track progress, and discuss issues in daily Scrum meetings. At the end of the sprint, the completed work is reviewed and feedback is collected for future improvements.
- The typical duration of a Sprint is **1 to 4 weeks**, with **2 weeks** being one of the most commonly used sprint durations in many software development organizations.

5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

Ans:

1. Backlog: -

The Backlog is a collection of all tasks, user stories, bug reports, and feature requests that need to be completed in the project. It serves as the master list of work items that can be selected for future sprints.

2. To Do: -

The To Do column contains tasks that have been selected from the backlog and are ready to be started by the development team. These tasks have not yet been worked on.

3. In Progress: -

The In Progress column contains tasks that are currently being developed, tested, or reviewed by team members. It shows that active work is being performed on those tasks.

4. Done: -

The Done column contains tasks that have been successfully completed, tested, and approved. These tasks no longer require any further work and are considered finished.

- Jira boards help software teams organize project activities, monitor the status of tasks, improve team collaboration, and track the overall progress of software development projects in an efficient manner.